DRAM Scheduling Policies

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DRAM Scheduling Policies

- FCFS (first come first served)
  - Oldest request first

- FR-FCFS (first ready, first come first served)
  1. Row-hit first
  2. Oldest first

  Goal: Maximize row buffer hit rate → maximize DRAM throughput

Solved Question

A DRAM main memory has 2 channels, 1 rank/channel.

It uses open-row policy. Latencies;

ACTIVATE: 15ns

Read/write: 15ns

Precharge: 15 ns

In each case, find stall time using i) an FCFS and ii) an FR-FCFS scheduling policies?

Row-buffer hit-latency = 15ns

Row-buffer conflict latency = 15+15+15 = 45ns

Case 1: Application "A" runs alone: FCFS

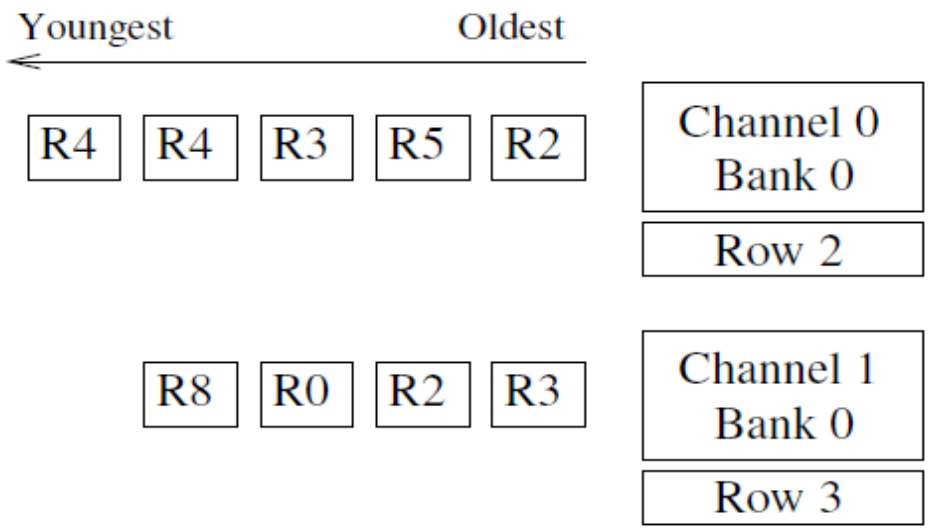
FCFS: Channel0: Two row-buffer hits (R2 and R4) and three row-buffer conflicts (R5, R3, R4).

Stall time = (15 *2) + (45 * 3) = 165 ns.

Channel1: One row-buffer hit (R3) and three conflicts (R2, R0, R8).

Stall time = (15 * 1) + (45 * 3) = 150 ns.

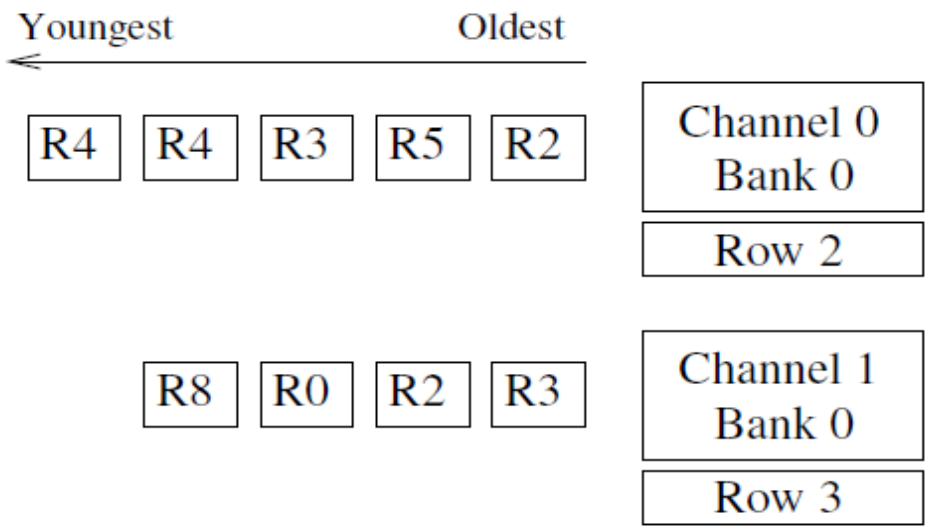
Overall stall time = max (150, 165) = 165ns



In bank 0 of channel 0: open row is Row2.
In bank 0 of channel 1: open row is Row 3.

Case 1: Application "A" runs alone: FR-FCFS

FR-FCFS: At both banks (and channels), the scheduling order is the exactly the same as when using FCFS, as the same requests hit in the row-buffer in both cases



Overall stall time = 165ns

In bank 0 of channel 0: open row is Row2.
In bank 0 of channel 1: open row is Row 3.

Case 2: Application "B" runs alone: FCFS

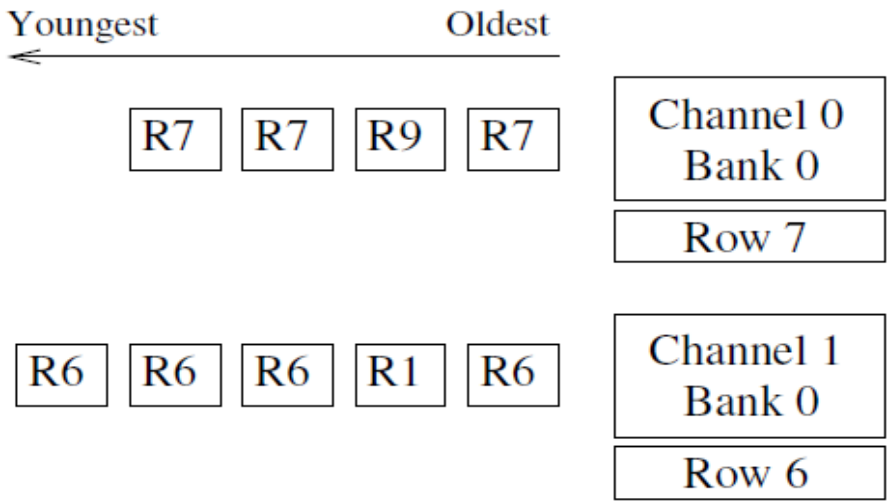
FCFS: Channel0: Two row-buffer hits (R7) and two row-buffer conflicts (R9 and R7).

Stall time = (15 *2) + (45 * 2) = 120 ns.

Channel1: Three row-buffer hit (R6) and two conflicts (R1, R6).

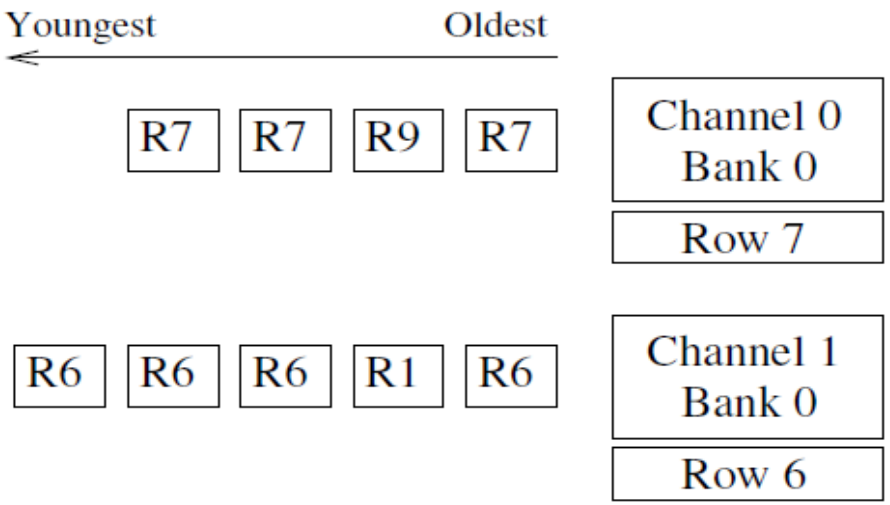
Stall time = (15 * 3) + (45 * 2) = 135 ns.

Overall stall time = max (120, 135) = 135ns

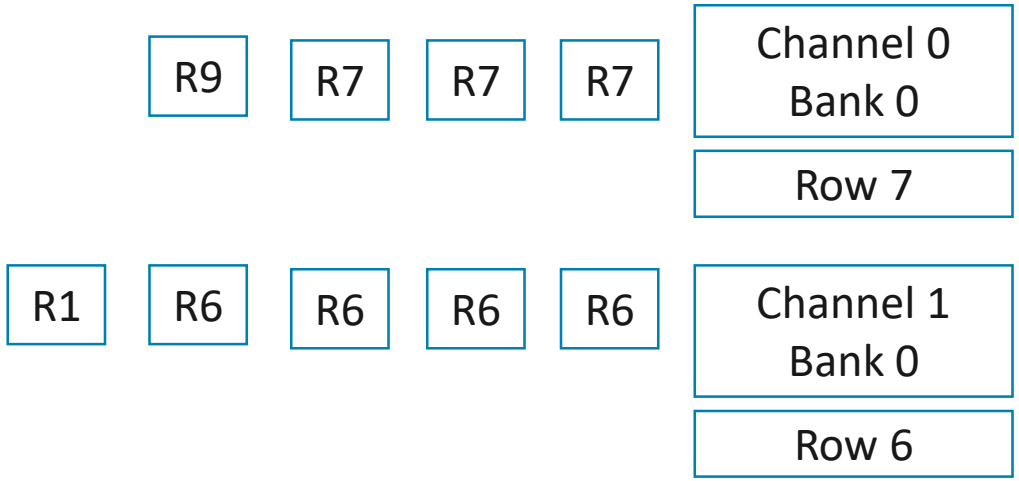


In bank 0 of channel 0: open row is Row7.
In bank 0 of channel 1: open row is Row 6.

Case 2: Application "B" runs alone: FR-FCFS



To improve row-buffer hits, we can reorder the requests as follows



Explanation: since R7 was open in channel0, we moved R7 ahead of R9 Since R6 was open in channel1, we moved R6 ahead of R1

Case 2: Application "B" runs alone: FR-FCFS

FR-FCFS: Channel0: Three row-buffer hits (R7) and one row-buffer conflict (R9).

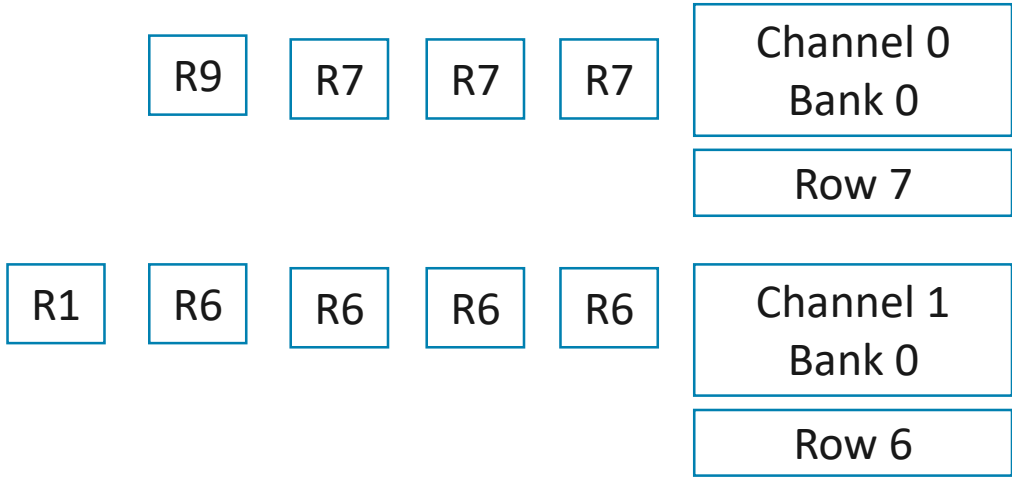
Stall time = (15 *3) + (45 * 1) = 90 ns.

Channel1: Four row-buffer hit (R6) and one conflict (R1).

Stall time = (15 * 4) + (45 * 1) = 105 ns.

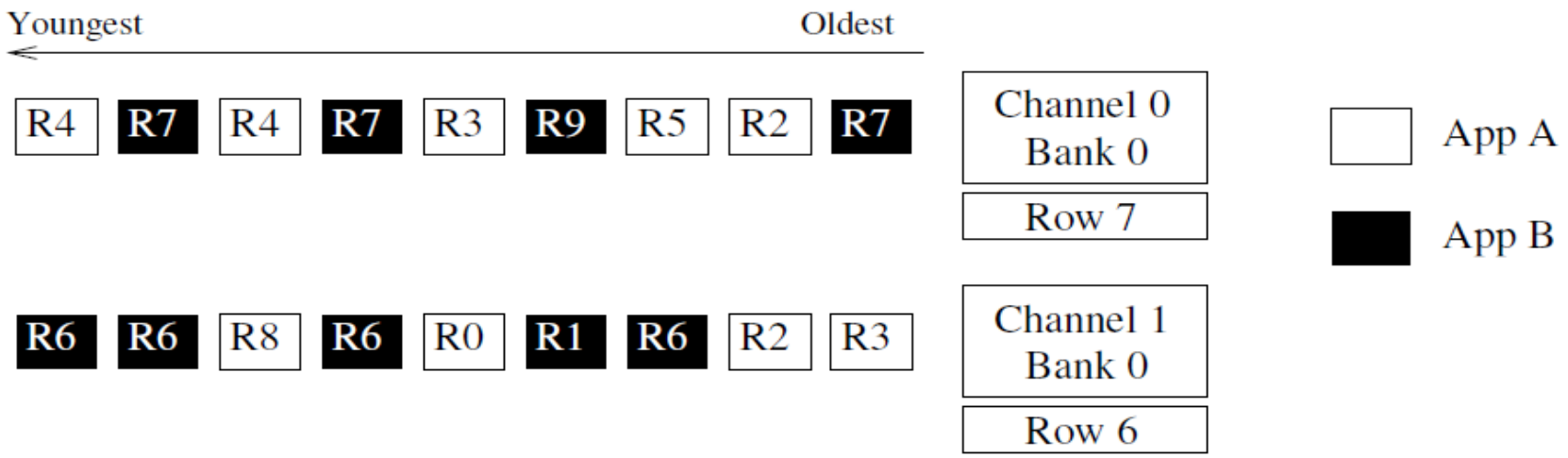
Overall stall time = max (90,105) = 105ns

Application B has higher row-buffer locality and hence benefits from FR-FCFS memory scheduling.



Snapshot after reordering

Case 3: Both A and B run.



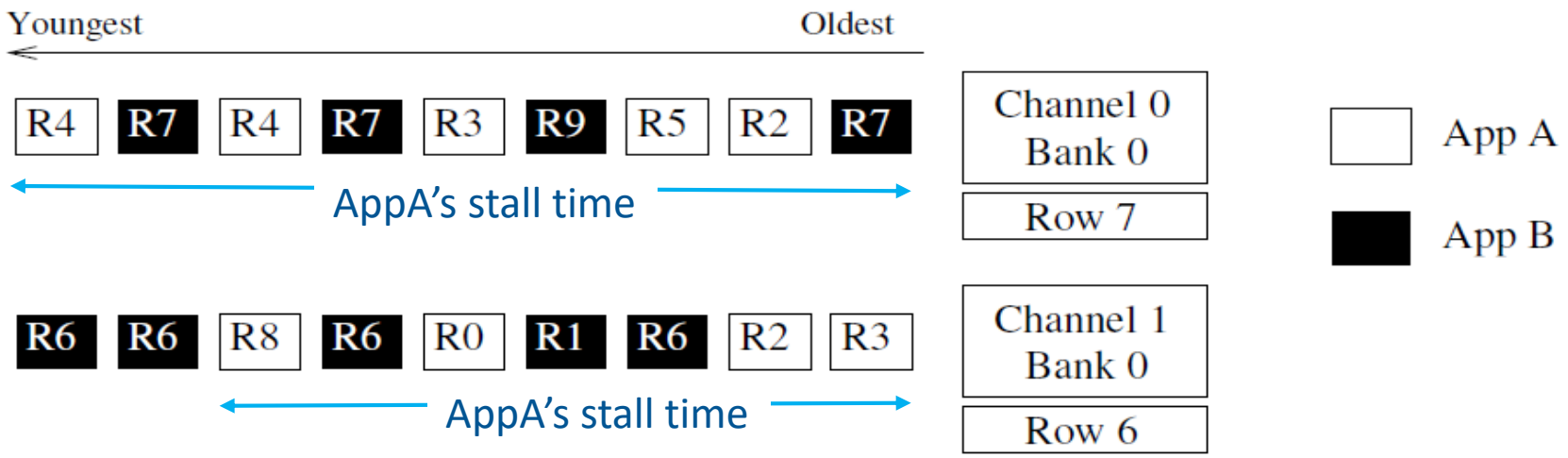
Find stall time of A, using FCFS and FR-FCFS

Find stall time of B, using FCFS and FR-FCFS

There is a single row-buffer, which is shared between two Apps.

When we ask stall time of A, we are asking time that requests of A get completed finally; not the sum of time the requests of A spend getting serviced.

Case 3: Both A and B run. FCFS, StallTime of A



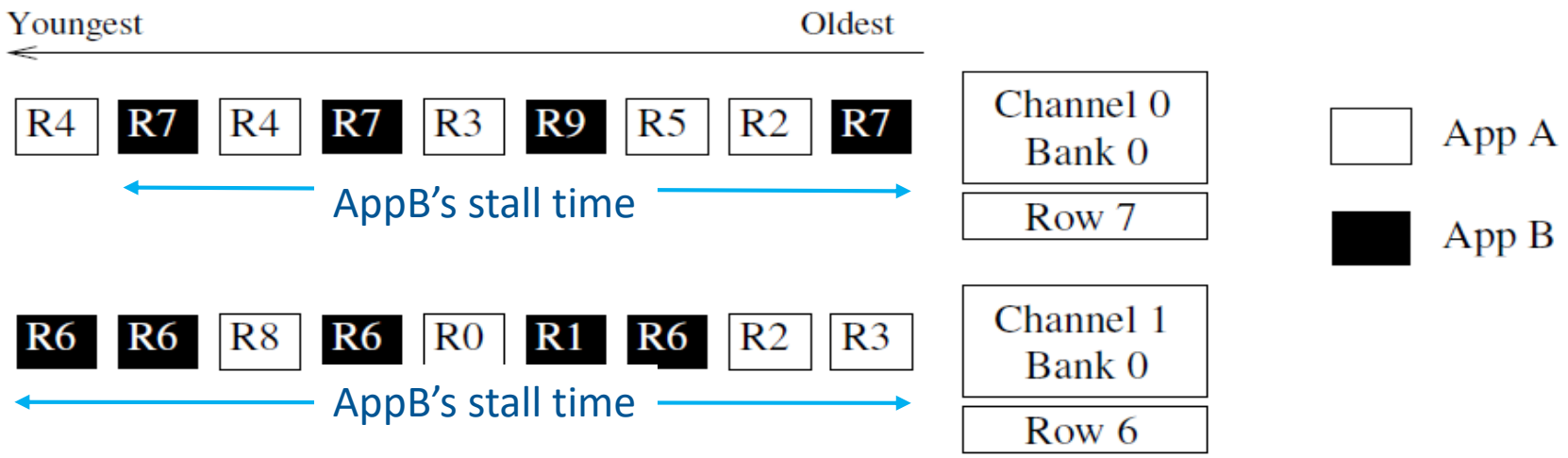
FCFS: Channel0: One row-buffer hits (R7) and 8 row-buffer conflict latencies (R2, R5, R9, R3, R7, R4, R7, R4). Stall time = 375 ns

Channel1: 7 row-buffer conflict latencies (R3, R2, R6, R1, R0, R6, R8).

Stall time at channel 1= 45 * 7 = 315ns.

Overall stall time = 375ns

Case 3: Both A and B run. FCFS, StallTime of B



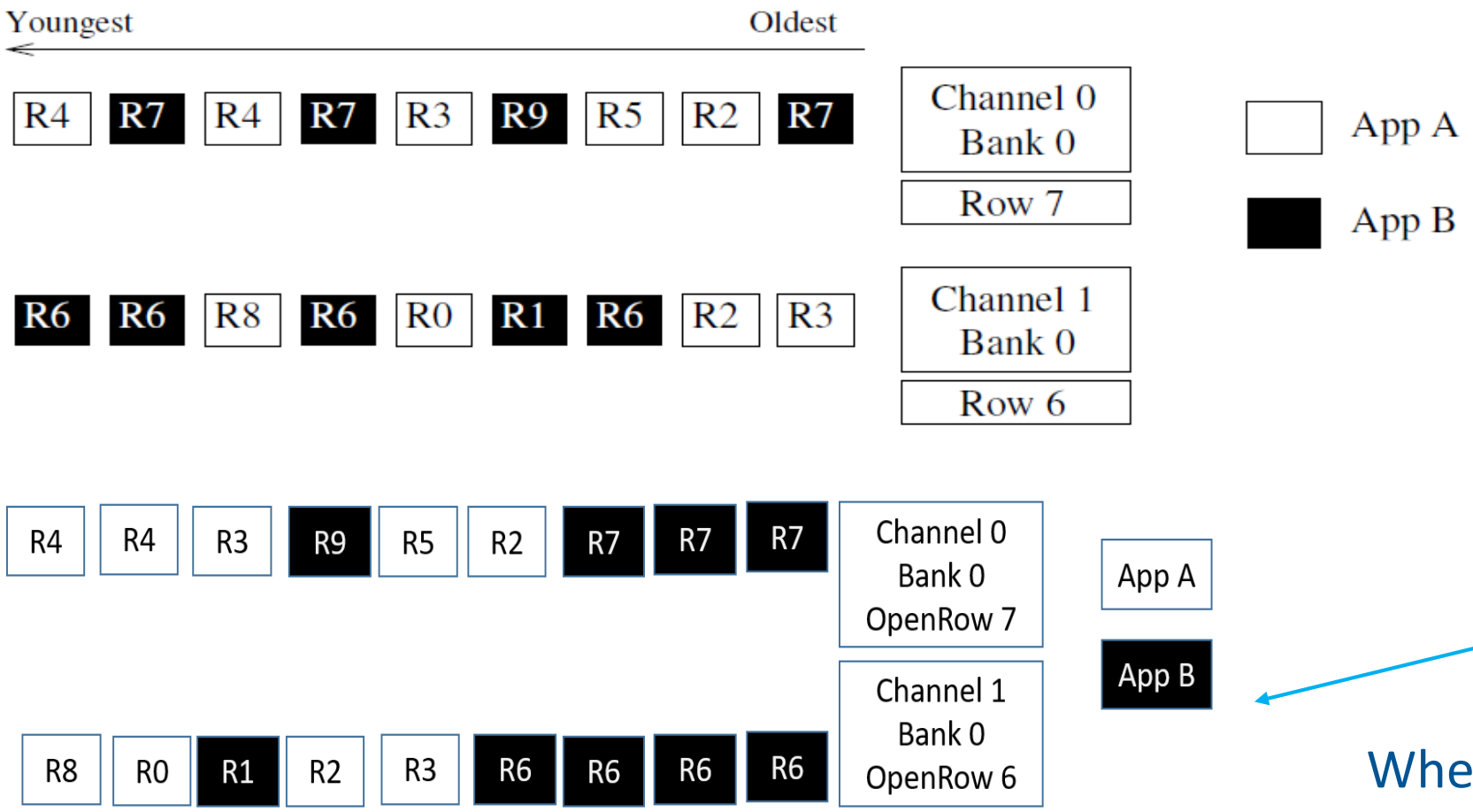
FCFS: Channel0: One row-buffer hits (R7) and 7 row-buffer conflict latencies (R2, R5, R9, R3, R7, R4, R7). Stall time = 330 ns

Channel1: 1 row-buffer hit and 8 row-buffer conflict latencies

Stall time at channel 1= 375ns.

Overall stall time = 375ns

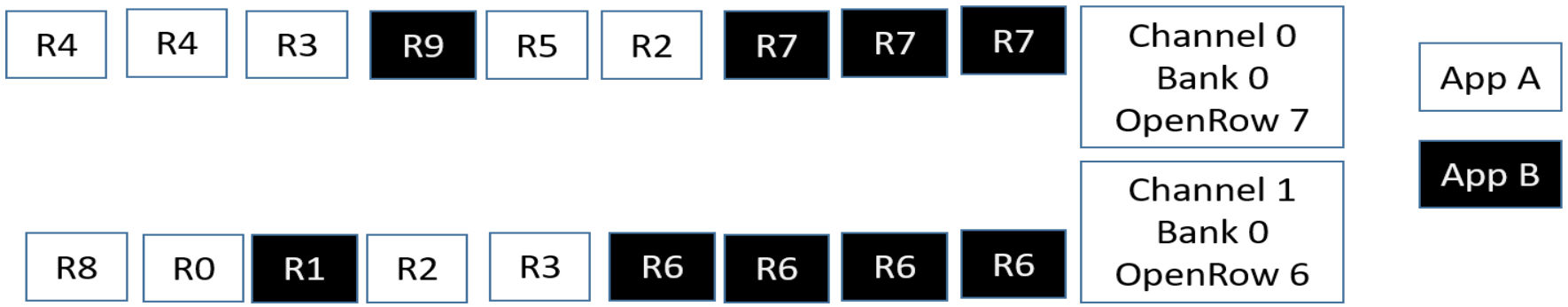
Case 3: Both A and B run. FR-FCFS



(a) Reordering of request done by FR-FCFS policy

To improve row-buffer hits, we can reorder the requests as follows

Whether the request is from A or B has NO impact on reordering.



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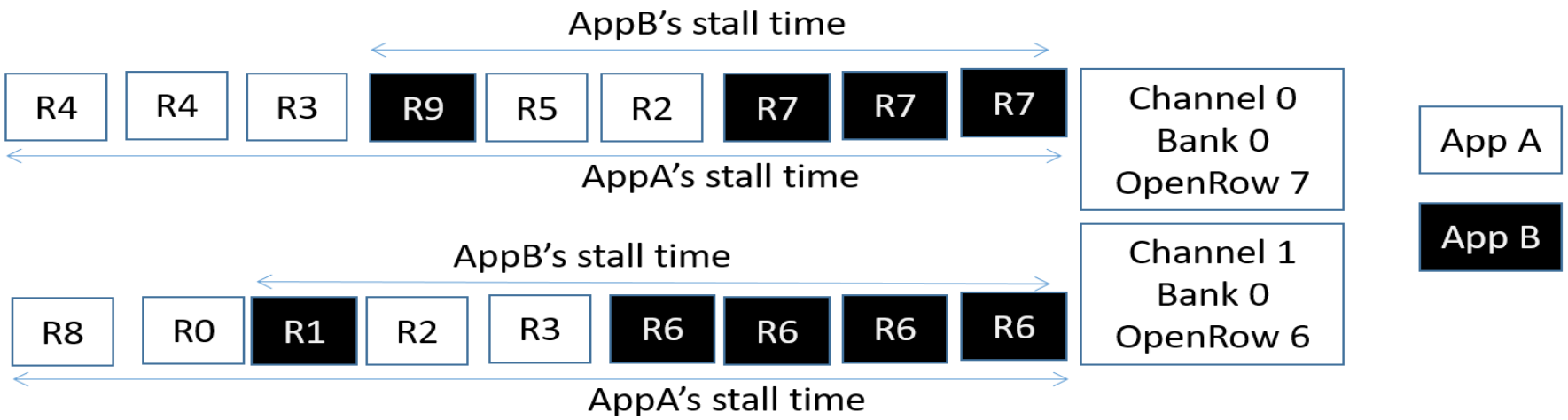


(b) Showing stall times on both channels for both apps in FR-FCFS policy

FR-FCFS, Stall time of A

We reorder them in the following manner (Also shown in Figure (a) above).

R4 R4 R3 R9 R5 R2 R7 R7 R7

At bank 0, channel 0, application A stalls for 4 row-buffer hit latencies (3 times to R7 and 1 time to R4) and 5 row-buffer conflict latencies (R2, R5, R9, R3, R4).

Stall time at channel 0 = (15 * 4) + (45 * 5) = 285ns.

At bank 0, channel 1, it stalls for 4 row-buffer hit latencies (R6) and 5 row-buffer conflict latencies (R3, R3, R1, R0, R8). This is because we will reorder the requests to improve row-buffer hits, we will reorder them in the following manner: R8 R0 R1 R2 R3 R6 R6 R6 R6

Stall time at channel 1 = (15 * 4) + (45 * 5) = 285ns.

Hence, stall time of application A is 285 ns.

FR-FCFS, Stall time of B

At bank 0, channel 0, 3 row-buffer hit latencies and 3 row-buffer conflict latencies.

Stall time at channel 0 = (15 * 3) + (45 * 3) = 180ns.

At bank 0, channel 1, 4 row-buffer hit latencies and 3 row-buffer conflict latencies.

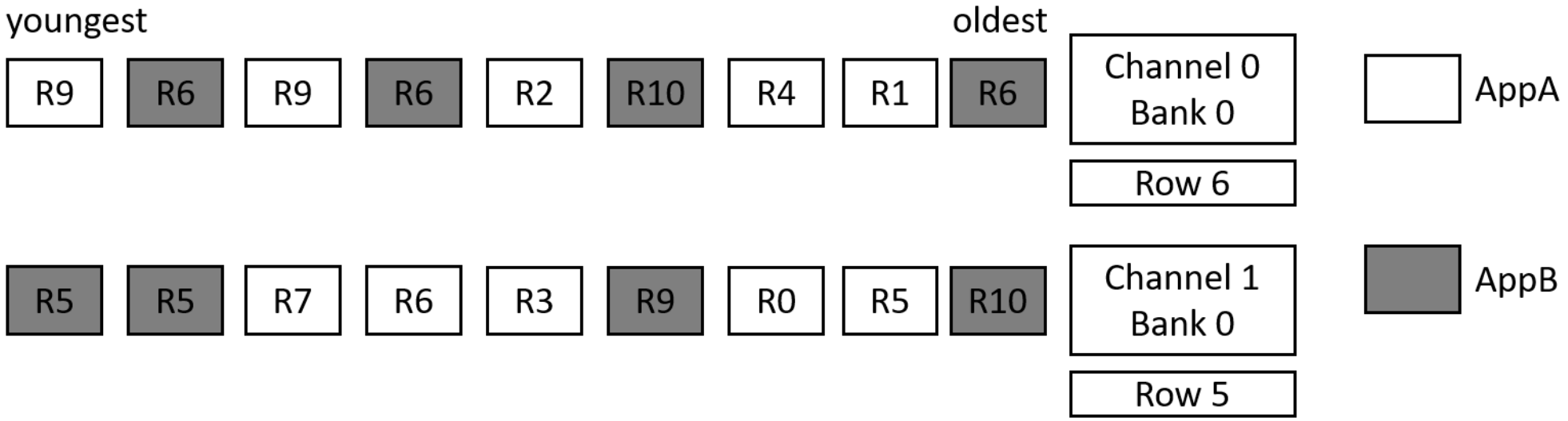
Stall time at channel 1 = (15 * 4) + (45 * 3) = 195ns.

Hence, stall time of application B is 195 ns.

Again note:, application B has higher row-buffer locality and hence benefits more from FR-FCFS scheduling.

Solved Question

In a DRAM, row-buffer hit, miss and conflict take 20, 35 and 65 ns latency. Find stall time of A and B, using FCFS and FR-FCFS.



Application A, FCFS:

Channel 0: 1 hit and 8 conflicts. Latency = 1*20+8*65 = 540ns

Channel 1: 7 conflicts. Latency = 7*65ns

So, stall time is 540ns.

Application A, FR-FCFS:

Channel 0: Accesses are reordered as: R6, R6, R6, R1, R4, R10, R2,

R9, R9 (old to new). Latency = 4 hits and 5 conflicts = 4*20+5*65 = 405ns

Channel 1: Accesses are reordered as: R5 R5 R5 R10 R0 R9 R3 R6

R7. Latency = 3*20+6*65 = 450ns

Stall time: 450ns

(c) Application B, FCFS:

Channel 0: 1 hit and 7 conflicts. Latency = 1*20+7*65 = 475ns.

Channel 1: 1 hit and 8 conflicts. Latency = 1*20+8*65 = 540 ns.

Stall time = 540ns.

(d) Application B, FR-FCFS:

Channel 0: access reordering is same as above shown for application A. But the difference is that it finishes earlier, when R10 is serviced. So, latency = 3*20+3*65 = 255ns.

Channel 1: B can finish when R9 is serviced. So, latency = 3*20+3*65 = 255ns.

Stall time = 255ns